Your Own Chatbot

The COSC 1210 capstone: one project instead of the weekly home assignments. You build a chatbot on the ELIZA engine and present it in class.

Marcus Birkenkrahe

COSC 1210 · CS I · Augustana University

Fall 2026

Meet SAGE, a study buddy

→ **Switch to Colab · `main()`**

```
SAGE: Hi, I am SAGE, your study buddy. What are you working on?
You: I'm nervous about the quiz.
SAGE: What part of it makes you nervous?
You: I don't understand regular expressions.
SAGE: Which part of regular expressions is unclear?
You: THIS IS NOT WORKING!
SAGE: No need to shout. What happened?
You: bye
SAGE: Good luck. Take a break before the next session.
```

SAGE is ELIZA with a different script. Your project: a bot of your own.

Engine and script

Engine cell – Python, given

```
def respond(sentence, script):
    sentence = normalize(sentence)
    for rule in script["rules"]:
        match = re.search(rule["pattern"],
                          sentence)
        if match is not None:
            ...
```

Persona cell – the script, yours

```
%%writefile sage.yaml
name: 'SAGE'
greeting: 'Hi, I am SAGE, your study buddy.'
rules:
  - pattern: '\bi don.?t understand (.*)'
    replies:
      - 'Which part of {x} is unclear?'
      - 'Can you explain {x} in one sentence?'
```

Same split as Weizenbaum's ELIZA (1966): one engine, many scripts.

Pick a track

1. **A new persona.** Your own script: a study buddy, a campus guide, a help desk, a cooking coach, a character you invent – plus one extension each. Most projects are this track.
2. **A bigger engine.** A persona with 8 rules instead of 12, plus two extensions each.
3. **Your own text program.** A text analyzer, a quiz program... only with an approved proposal.

Extensions: your own code in the engine

Each of you adds one extension (track 2: two), written as new functions with docstrings and 2 doctests each. Pick one, or propose your own:

- **Memory**: the bot keeps your "my ..." phrases and brings one up later – "Earlier you said your sister..."
- **Statistics**: count which rules fire; at the end, print the counts and draw a bar chart
- **Your name**: the bot asks your name and uses it in replies, `{name}` in the script
- **Tone**: capitals and questions get their own replies
- **Markov fallback**: a reply built from your own earlier sentences (chapter 12)

What you hand in

- one Colab notebook, from the starter: *Run all* runs without an error
- your persona `yourname.yaml`: at least 12 rules with 2 replies each; 4 use `{x}`, 3 use `\b`; your own greeting, goodbye, fallback, reflections
- your extension(s): new functions, each with a docstring and 2 doctests
- a test conversation: 15 lines that fire every rule, and its transcript
- AI-use statement in the first cell: tool, what for, what you checked

Milestones

Milestone	Due	Points	Hours, each
Proposal	Mon 11/09, 2:00 pm	3	1–2
Prototype	Mon 11/23, 2:00 pm	7	3–4
Final code	Tue 12/01, 11:59 pm	10	3–4
Presentation	in class, 12/02 or 12/04	10	1
Total		30	8–11

In a pair: 8–11 hours each, 16–22 for the team – about 2–3 hours a week each.
Alone: 10–14 hours, about 3 hours a week.
No late submissions: Canvas closes at each deadline.

The presentation: 6 minutes + 3 for questions

1. What your bot is for
2. Live demo: your test lines, then two or three lines from the audience
3. One rule: its pattern, and why it is above or below another rule
4. Your extension, explained by the person who wrote it
5. One thing that went wrong and how you fixed it
6. How you used AI and what you checked

Present from your notebook or from slides (at most 3); upload what you present: `.ipynb` or PDF.

Alone or in a pair

- A pair is recommended: two of you, one bot, one presentation
- Each of you writes your own extension and explains it
- Both of you speak in the presentation
- Rules for every persona: no real people, no medical, legal or crisis advice, replies you would show in class

What you need, and where it comes from

You need	You get it
patterns: `re.search`, groups, `\b`	the next lecture
lists, dictionaries	the next two weeks
YAML files	after dictionaries

Everything runs in Colab: nothing to install.

Next steps

1. Open the starter notebook, run all, talk to DOCTOR: `https://birkenkrahe.github.io/go/capstone`
2. Find a partner, or decide to work alone
3. Think of a persona: who would talk to your bot, and about what?
4. Write five sample exchanges – that is most of your proposal

The full brief is on Canvas.